

around 0.43. Thus there are some long-horizon common components in NCREIF and REIT returns. Econometric studies also document strong evidence that public real estate markets lead private real estate markets, which is to be expected since price discovery should happen in markets that are liquid.³⁶

The astute reader will be wondering at this point, if direct and indirect real estate do move together in the long run, is real estate a separate asset class given that the correlation between REITs and equities is so high (63% in Table 11.11)? CPPIB, the manager of the CAN\$162 billion Canada Pension Plan, does not consider real estate a separate asset class; CPPIB practices factor investing when it invests in real estate and thinks of real estate as comprising equity and debt characteristics.³⁷ In chapter 10, we also found that real estate did not seem to yield additional returns in excess of stocks, bonds, and REITs in factor benchmarks. But CPPIB is an exception: most investors regard real estate to be a separate asset class (or even factor). REITs cannot be the basis for this statement, given the high correlation of REIT returns with equities in Table 11.11. The statement is usually made using direct real estate returns—but as we know, direct real estate returns are not returns.³⁸ Although REIT and NCREIF returns do exhibit some commonality in the long run, the long-horizon correlations are far from one as Figure 11.12 shows. Thus:

Direct and Indirect Real Estate Are Different

Listed REITs seem to have one major advantage over direct real estate: they are liquid. But as of June 30, 2012, there were only 150 REITs traded on the New York Stock Exchange with a collective market capitalization of \$545 billion. This is an average market capitalization of \$3.6 billion for each REIT—smaller than many big real estate developments in the world’s leading cities. The FTSE NAREIT All Equity REITs Index had a total market capitalization of \$562 billion. In contrast, Florance et al. (2010) estimate that the (direct investment) commercial real estate market in the United States totaled over \$9 trillion at the end of 2009, even though the market lost over \$4 trillion from 2006 to 2010.

If REITs have such a big liquidity advantage, why is the amount of money invested through REITs well below one-tenth of the total commercial real estate market? Why aren’t REITs more popular?

Structurally, REITs are different from direct real estate. Prior to the early 1990s, the REIT industry was very small because REITs were forced to retain external advisors to select and execute investment strategies. In 1986, REITs were allowed to select and manage their own assets and during the early 1990s many

³⁶ See Gyourko and Keim (1992), Barkham and Geltner (1995), and Oikarinen, Hoesli, and Serrano (2011).

³⁷ See the case “Factor Investing: The Reference Portfolio and Canada Pension Plan Investment Board,” Columbia CaseWorks #120302. Assets under management as of March 31, 2012.

³⁸ See Mei and Lei (1994) and Ling and Naranjo (1997, 1999).